

Mosaic Effect Risk and Mitigation

Technical Report for IT and Research Operations — SONA / PRAMS

People Analytics Lab of the Bayou

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i How to render (maintainers)

From the repository root (with [Quarto](#) installed):

```
./venv312/bin/python scripts/visualize_mosaic_simulation.py
quarto render docs/mosaic_effect_technical_report.qmd
quarto render docs/mosaic_effect_technical_report.qmd --to pdf
```

Figures: docs/figures/ · Interactive charts: [mosaic_simulation_charts.html](#)

1 Abbreviations

This report uses the following abbreviations:

Abbreviation	Meaning
API	Application programming interface
BF	Bayes factor
CSRF	Cross-site request forgery
DLP	Data loss prevention
DMP	Data management plan
DUA	Data-use agreement
FERPA	Family Educational Rights and Privacy Act
IDOR	Insecure direct object reference
IRB	Institutional review board
IT	Information technology
MFA	Multi-factor authentication
NTP	Network Time Protocol
OSF	Open Science Framework
PII	Personally identifiable information
QI	Quasi-identifier
QC	Quality control
RBAC	Role-based access control
SDC	Statistical disclosure control
TOP	Transparency and Openness Promotion
WAF	Web application firewall

2 Executive summary

The SONA / PRAMS platform collects protocol responses as flexible JSON while operating in a FERPA-aware, IRB-governed environment. For studies with sample sizes between **20 and 500**, the intended public disclosure path is **not row-level microdata** but **aggregates and inferential statistics** (means, proportions, confidence intervals, Bayes factors, and test statistics) that support scientific claims. That choice materially lowers re-identification risk, yet it does not eliminate the **mosaic effect**: the risk that a person becomes identifiable when quasi-identifiers, behavioral data, and auxiliary records (signups, emails, timing, external identifiers) are combined across systems or releases.

This report is written for two audiences who share accountability. **IT professionals** must understand where identifiers and metadata enter the stack, how exports and backups propagate them, and which access controls bound insider and pipeline risk. **Research operations** must apply **statistical disclosure control (SDC)** so that internal analysis files and any exceptional microdata handoffs do not undermine the aggregate-first publication strategy. A Monte Carlo simulation (§4.5) quantifies how often participants are unique on realistic quasi-identifiers at $N = 20\text{--}500$ and supports a practical release rule: **external handoffs should be descriptive-statistics tables unless the study is very large relative to the fields collected**. The diagrams below summarize data flow and disclosure tiers; later sections map controls to concrete models and code paths in this repository.

Parameter	Value
Typical study N	20–500
Primary external release	Aggregates and inferential statistics
Raw microdata release	Unlikely; not the default path
Institutional frame	FERPA-aware recruitment; IRB human subjects governance

3 Document standards

This report follows practices from the [K-Dense scientific-agent-skills](#) collection:

Skill	Application here
markdown-mermaid-writing	Architecture and workflow diagrams are Mermaid in Quarto (version-controlled, diffable source)
scientific-writing	Technical prose structure, precision, and tables for comparisons (adapted for IT/research ops, not journal IMRAD)
Statistical disclosure control	k -anonymity framing and cell-size rules aligned with social-science aggregate release

4 1. Problem statement: the mosaic effect

4.1 1.1 Definition

The **mosaic effect** (also discussed under **statistical disclosure control** and **record linkage** in federal and health-data guidance) arises when no single field identifies a person, but **joint information** does. In this codebase, quasi-identifiers (QIs) include demographics in JSON payloads, submission timestamps, session identifiers, vignette presentation order, and professional attributes in HR-SJT instruments. Auxiliary data may include course rosters, **Signup** timeslots, **StudyEmailContact** rows, or PRAMS **participant_secure_id** values held outside the anonymous-looking **Response** table.

Risk increases when the equivalence class size k for a combination of QIs falls below an acceptable threshold (k -anonymity). With N between 20 and 500, even two or three QIs can yield **singleton cells** in classroom cohorts or rare demographic intersections.

4.2 1.2 Aggregate release is appropriate—and not sufficient alone

Publishing group-level statistics is much safer than releasing rows, but it is not a free pass. Small cells in slides, unusual response patterns in appendices, and linkage across repeated OSF deposits can still expose people.

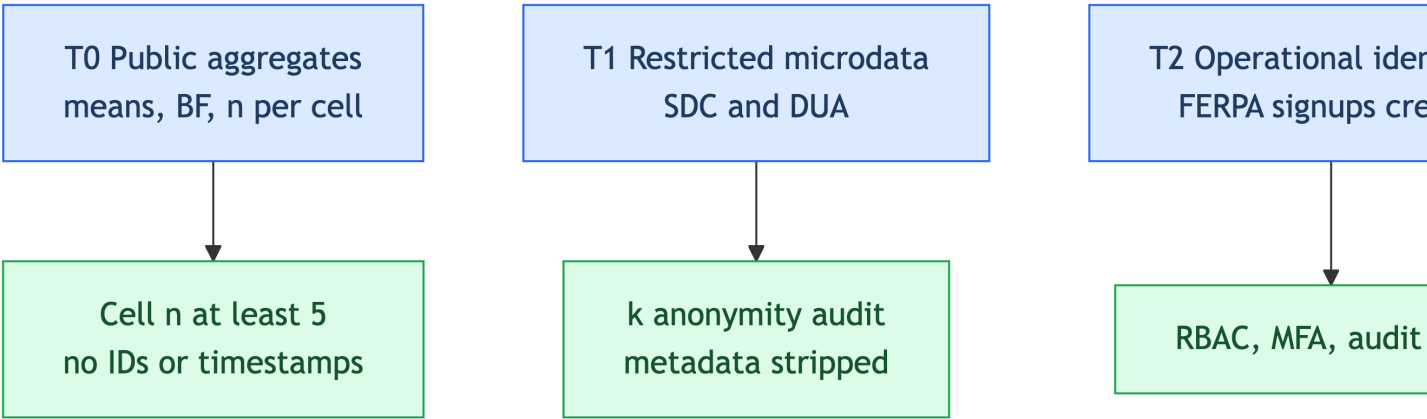
Internal CSV/R exports are another weak spot because they often sit outside the publication workflow and get less scrutiny. Since raw release is rare in this lab, the practical focus is straightforward: keep row-level data from drifting out of the application, and apply cell-size and suppression rules to every table that leaves the secure environment.

4.2.1 Open science without mosaic risk

Open science is not an all-or-nothing obligation to post everything (Castille et al., 2022). The “buffet” framing in that editorial is useful here: preregistration, materials sharing, code sharing, and data sharing are different choices, not one package deal. For studies in the 20-500 range, the best fit is transparent design and analysis plus descriptive tables on the Open Science Framework (OSF). That keeps the work reproducible without turning internal response JSON into public microdata by accident.

Scientific value (Castille et al., 2022)	Open-science practice	Mosaic-safe interpretation for this lab
Openness	Preregistration, OSF project, shared materials	Document instruments and analysis; deposit aggregates and code , not identifiable rows
Honesty	Report what was collected and what is shared	Data management plan (DMP) states QIs collected; OSF README states T0-only external data
Stewardship	Protect participants and institutional relationships	FERPA signups stay T2; OSF gets tables with cell $n \geq 5$, not session IDs or exact age
Accountability	Reproducible workflow	Code + summary tables reproduce claims; microdata stays behind DUA if needed

OSF deposits, conference slides, and supplements are publication channels, so they need the same disclosure rules as journal tables. A preregistered hypothesis is not a reason to post a cross-tab with singleton cells. Materials can be fully open while data remain aggregate-first. That still aligns with Transparency and Openness Promotion (TOP) guideline goals, and it avoids treating “open data” as “open rows.”



Disclosure tiers and typical controls (T0 is the default external path for this lab).

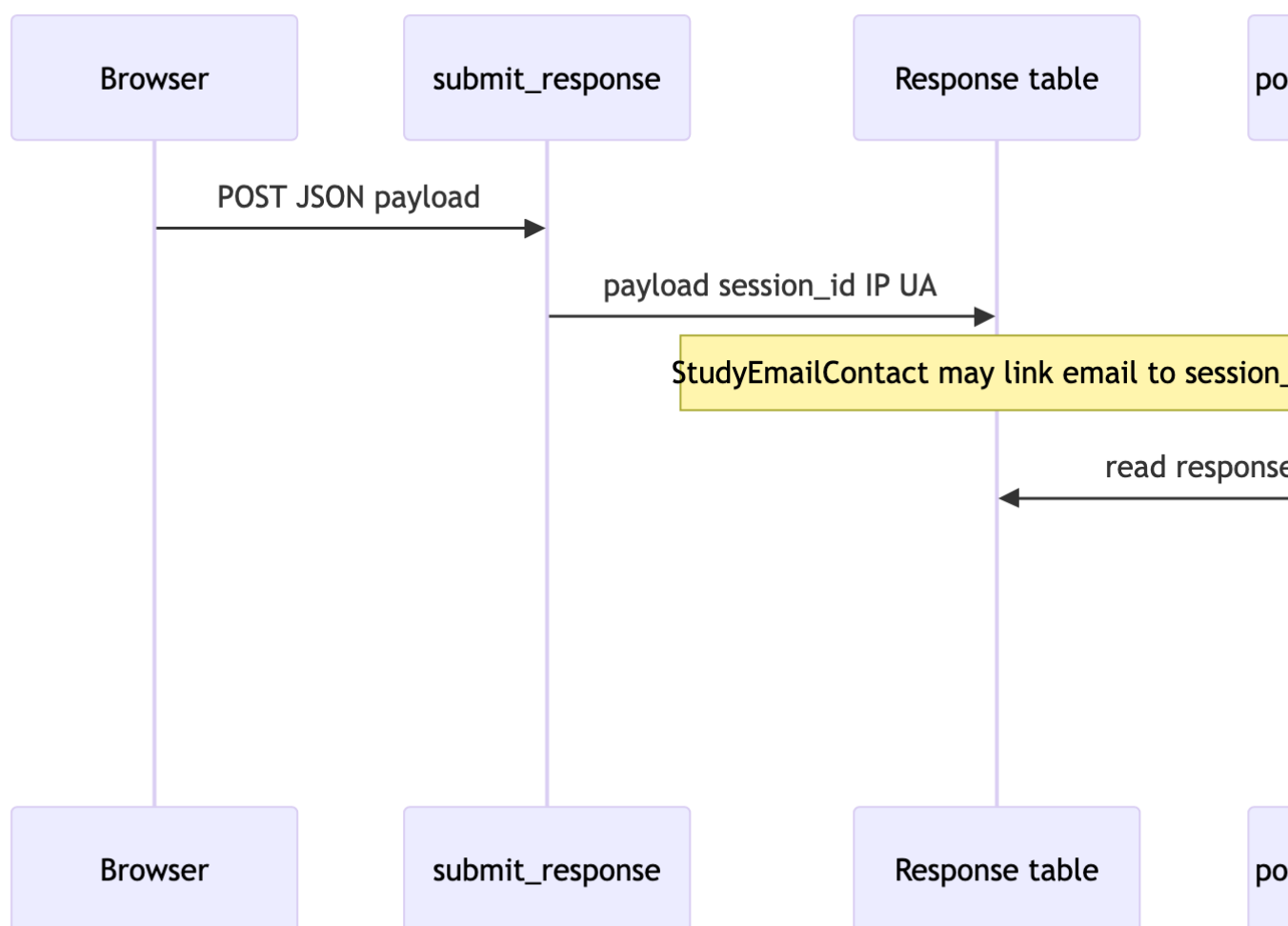
5 2. System architecture and data flow

5.1 2.1 Entity model (what IT inherits)

Store / model	Role	Identifier posture
Response	payload JSON + session_id, created_at, ip_address, user_agent	No participant FK; anonymity depends on payload
Signup	Timeslot booking linked to User	Identified (FERPA-relevant)
StudyEmailContact	Infographic email + optional session_id	Bridge to Response
StudentDataConsent	Secondary-use consent with email, names	Identified
PrescreenResponse	Participant- linked prescreen JSON	Identified
PRAMSSignup	participant_session_id, cancellation PIN	Pseudo-anonymous in PRAMS; may link externally
AuditLog	Actor, action, metadata, IP, user agent	Security and compliance

The **Response** model is anonymous by schema design (no participant foreign key). **Client payloads and cross-table joins can restore identity** if governance is weak.

5.2 2.2 End-to-end flow



Browser data flow from submission through storage, analysis export, and publication.

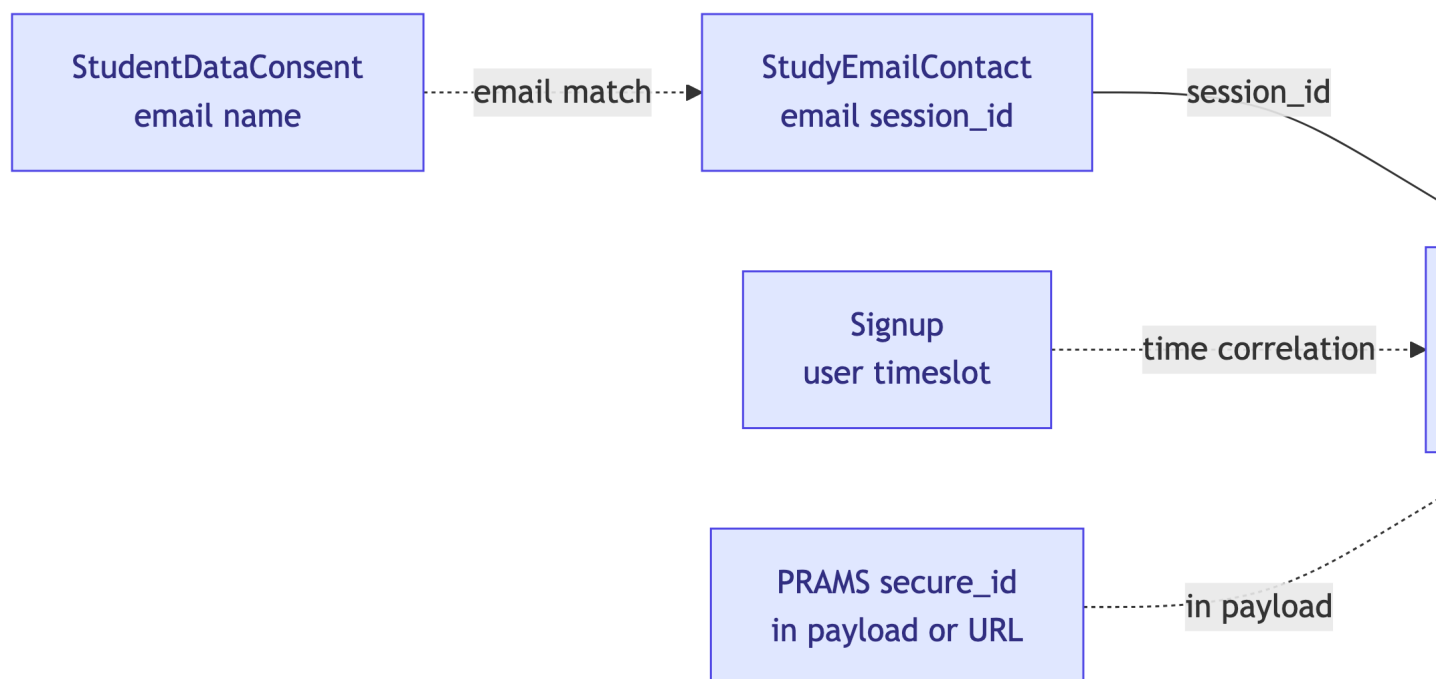
Ingestion (`apps/studies/views.py`, `submit_response`): full client JSON is persisted without field-level allowlisting; `REMOTE_ADDR` and `HTTP_USER_AGENT` are stored (HR-SJT demographics use `get_client_ip` with `X-Forwarded-For`—proxy configuration should be consistent).

High-risk payload fields (goals-refs live survey): exact `demographics.age`, `demographics.gender`, `demographics.student_status`, `prams_participant_id`, `completed_at`, `results_email`, `vignette_order_presented`, `goal_frames`, client `user_agent`.

Export (`apps/studies/tasks.py`): `run_post_decision_analysis` writes a temporary CSV with `response_id`, `session_id`, `created_at`, and serialized payload for Rscript. The file is deleted in `finally`, but the R process or analyst environment may retain copies.

Pseudonymization utility: `config/export_utils.py` provides `get_anonymized_participant_id()` (HMAC with `PARTICIPANT_EXPORT_SALT`) but is **not wired** into the main response export path—IT should not assume automatic HMAC IDs on export.

5.3 2.3 Mosaic linkage scenarios (this system)



Cross-table linkage paths that can re-identify anonymous-looking responses.

6 3. Key concerns for IT professionals

IT ownership centers on **confidentiality, integrity, and auditability** of systems that touch participant data—not on choosing statistical tables for papers.

6.1 3.1 Security and access control

Concern	Description	Posture in this codebase
Authentication boundary	Who may POST protocol data	<code>submit_response</code> is unauthenticated; gated by <code>Study.active_approved</code>
RBAC	Participant / researcher / IRB / admin	Role model present; see <code>SECURITY_COMPLIANCE_REMEDIATION_LIST.md</code>
IDOR	Object-level authorization	Documented remediation for attendance and similar views
CSRF	Cross-site POST to study APIs	Requires explicit token strategy per deployment
Admin surface	Django admin shows full payload, IP, UA	High-trust accounts only; MFA

6.2 3.2 Data classification and FERPA

Enrollments, credits, signups, and prescreens linked to **User** are **education records** when maintained by the institution. **Response** rows may fall outside FERPA only while they remain **non-linkable**; linkage via email, session, or timing collapses that boundary. Secondary-use consent (**StudentDataConsent**) is a

separate legal track from anonymous protocol JSON. IRB AI review may transmit protocol text to external LLMs—see `FERPA_AUDIT_REPORT.md` for prompt-screening gaps.

6.3 3.3 Infrastructure and pipeline leakage

Mechanism	IT action
<code>ip_address</code> , <code>user_agent</code> on Response	Restrict column access; exclude from backups shared with researchers
Precise timestamps	Treat as QI in exports; align NTP and proxy logging
UUID keys in CSV/R temp files	Encrypt analyst workstations; no sync to personal cloud
DB backups, WAF, app logs	Retention policy; DLP on export endpoints
<code>course_credits_csv</code> (FERPA)	Log access (implemented); separate storage from research

6.4 3.4 Monitoring and audit

`AuditLog` records sensitive actions; signal-created entries **lack IP/user agent** (`apps/studies/signals.py`). IT should centralize logs, review admin access periodically, and alert on anomalous `Response` creation rates (data poisoning).

6.5 3.5 What IT should expect from research (aggregate-first)

When research affirms that only **T0** outputs leave the institution, IT still validates that **T1** files are not stored on open shares. Research attestation should reference cell-size rules (§4.1) and the audit script (§5).

7 4. Research-side mitigation (statistical disclosure control)

Research operations implement SDC complementary to IT controls. Because external release is **aggregate-first**, the highest-impact research practices are **small-cell suppression**, **QI minimization**, and **internal microdata hygiene**.

7.1 4.1 Cell-size and k rules for $N = 20\text{--}500$

N range	Minimum reported cell n	Internal k review (if microdata exists)
20–49	5 (collapse or suppress otherwise)	Target k 3–5; avoid 3+ way demos in tables
50–199	5	Target k 5
200–500	5 (10 for rare HR/professional categories)	Target k 5–10

For journals, OSF, and conference slides: **do not display cells with $n < 5$** . For inferential claims, report test statistics and effect sizes at the **condition or scale level**, not per-participant series, unless n is large and points are aggregated.

7.2 4.2 Collection-time mitigations

Harmonize instruments so live surveys match lower-risk patterns already used in `ext_am4_assessment.js` (age **bands**, not exact age). Route optional emails through `StudyEmailContact`, not `results_email` inside

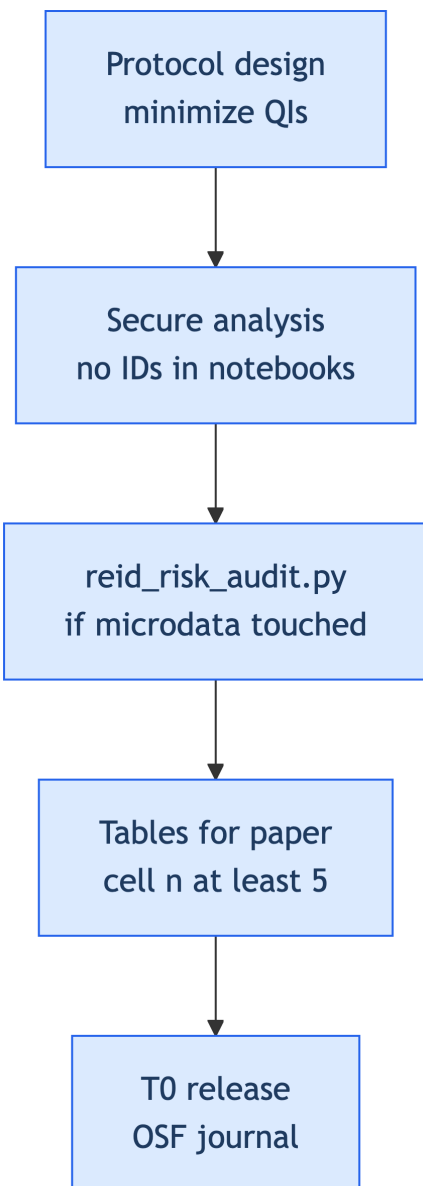
payload. Avoid echoing PRAMS participant IDs into JSON unless IRB-approved linkage is required; if required, store keys in a **locked linkage table**, not in analysis copies. Omit client `user_agent` and ISO `completed_at` from payload when server `created_at` suffices for operations.

7.3 4.3 Analysis-time mitigations

Researchers should flatten `payload` for internal QC, run the k-anonymity audit (§5), coarsen age and dates, top/bottom-code numeric outliers, apply local suppression on sparse cells, and compute inferential results inside the secure environment. **Only summary tables** graduate to T0.

7.4 4.4 Publication and OSF

Preregister the disclosure tier in the protocol and data management plan. Suppress interaction cells with low n . Do not publish vignette order vectors with demographics unless aggregated. Separate FERPA operational exports from analysis directories on storage.



Recommended research workflow from collection through publication (aggregate-first).

7.5 4.5 Monte Carlo simulation: plain-language results

A population-level simulation (`scripts/mosaic_reidentification_simulation.py`) draws many random samples at sizes $N = 20$ -500 and asks a simple question: how often does one person end up alone in a quasi-identifier cell? Here that means being the only row with a given combination such as gender, exact age, and student status. The figures label that rate as $P(\text{unique})$. In plain terms, it is an estimate of how easy row-level matching can become when detailed microdata exist internally, even if nobody plans to publish those rows.

How to read the charts: each line reflects a different level of detail in the collected fields. When $P(\text{unique})$ is high, those fields start behaving like a partial identifier. If someone already knows a student's age, gender, and status, they can often find one matching row. The second student chart tracks a different threshold: the share of participants in cells smaller than five, which is the common minimum for tables in papers and slides. The HR charts add career attributes (race, job level, years in HR, credentials), and uniqueness rises fast. The linkage chart models a partial roster attack. The mitigation bars compare three policy options at $N = 100$.

Finding (student, gender + exact age + status, no mitigation)	Approximate value
$P(\text{unique})$ at $N = 20$	56%
$P(\text{unique})$ at $N = 100$	18%
$P(\text{unique})$ at $N = 100$ after age bands	4%
$P(\text{unique})$ at $N = 100$ after suppressing cells with $k < 5$	0%
Share of sample in cells with $k < 5$ at $N = 100$	55%
HR four-way QIs, $P(\text{unique})$ at $N = 50$	68%
Roster linkage success at $N = 100$	4–5%

Student cohort: exact age creates a lot of uniqueness on its own at classroom sample sizes, and adding student status makes that worse. At $N = 100$, roughly one in six participants is unique on gender + exact age + status. At $N = 20$, it is more than half. Switching to age bands cuts the $N = 100$ result to about one in twenty-five. Suppression of sparse cells (or collecting less granular fields from the start) can remove singleton cells in this scenario. Even with no row-level public release, the below- k curve shows how easy it is for small cross-tabs to leak identity.

HR cohort: once you combine several career attributes, uniqueness climbs quickly. With four QIs at $N = 50$, about two-thirds of participants are unique. The rate drops as N increases, but it is still non-trivial at $N = 500$ for the richest field combination.

Linkage and mitigations: roster linkage is a secondary path compared with in-sample uniqueness, but in this simplified model it still sits around 4-5% near $N = 100$. The mitigation comparison at $N = 100$ is the practical takeaway: coarsening helps, and suppression (or tighter collection rules) can bring both $P(\text{unique})$ and small-cell share to zero in this case.

Full numeric output: `docs/mosaic_simulation_results.csv`. Regenerate figures with `scripts/visualize_mosaic_simulation.py`.

7.5.1 Recommended external release format

For studies in the 20-500 participant range, the safest default is simple: what leaves the secure environment should be descriptive tables (or a small set of tables), not row-level files and not high-dimensional tabs with thin cells. Inferential outputs can still be reported from secure analysis (test statistics, Bayes factors, confidence intervals). The key boundary is that external files should not be reversible to a person.

That can change when datasets are very large relative to the identifiers collected (for example, thousands of rows with coarse demographics), because row-level uniqueness drops. In this lab, that is the exception, not the norm. Operationally, treat Celery-to-R CSVs, notebook extracts, and OSF uploads as T0 summaries

only. If a collaborator needs more detail, use a data-use agreement, disclosure controls, and the audit script in §5 instead of sharing raw exports by email.

! Release checklist (research PI)

1. External handoff = **means, proportions, SDs, test stats, Bayes factors, cell n** —not participant-level rows.
2. Every published cell has **$n \geq 5$** (collapse or suppress otherwise).
3. No exact age, email, session ID, vignette order vectors, or timestamps in public materials.
4. Run `reid_risk_audit.py` before any exceptional microdata share when $N < 200$ or 3 QIs are collected.

7.5.2 Simulation figures

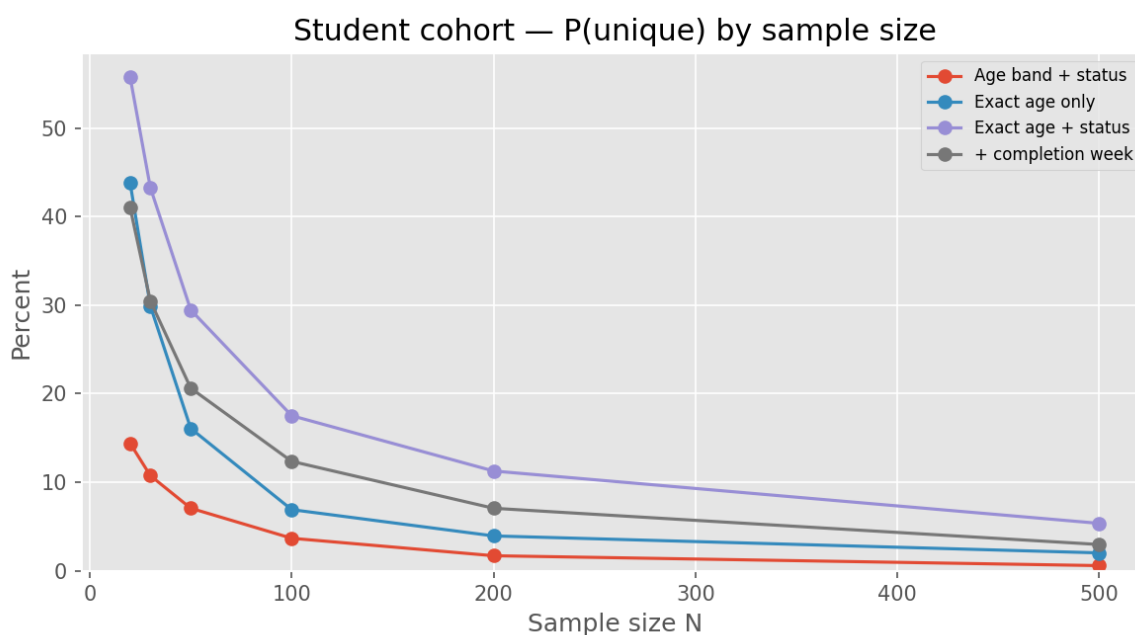


Figure 1: Student cohort: probability a participant is alone in their QI cell ($P(\text{unique})$) by sample size.

8 5. Operational tooling

Repository script: `scripts/reid_risk_audit.py` (pandas). Run after exporting responses from Django (management command or admin export), **before** any collaborator handoff or R analysis on row-level data.

```
# Risk report (aggregate release: still useful for internal QC)
```

```
python3 scripts/reid_risk_audit.py \  
  --input responses_export.csv \  
  --flatten-payload \  
  --k-threshold 5 \  
  --max-combo 3 \  
  --strip-identifiers \  
  --output-risk k_risk_report.csv \  
  --output-summary-json k_risk_summary.json
```



Figure 2: Student cohort: share of participants in equivalence classes smaller than the publication threshold $K = 5$.

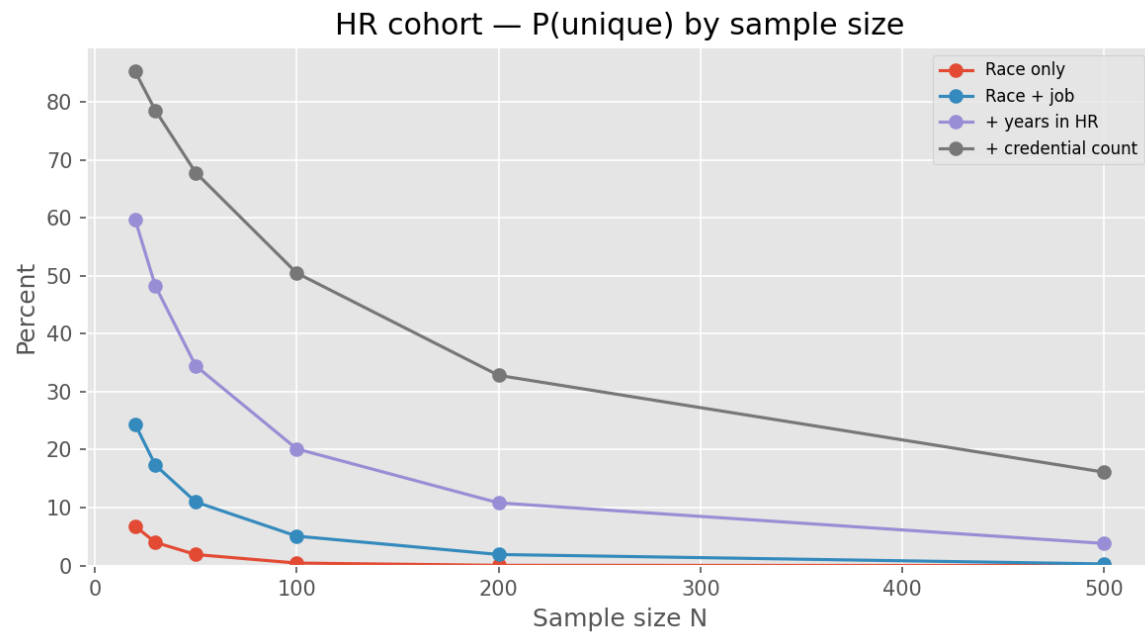


Figure 3: HR professional cohort: $P(\text{unique})$ by sample size as quasi-identifiers are added.

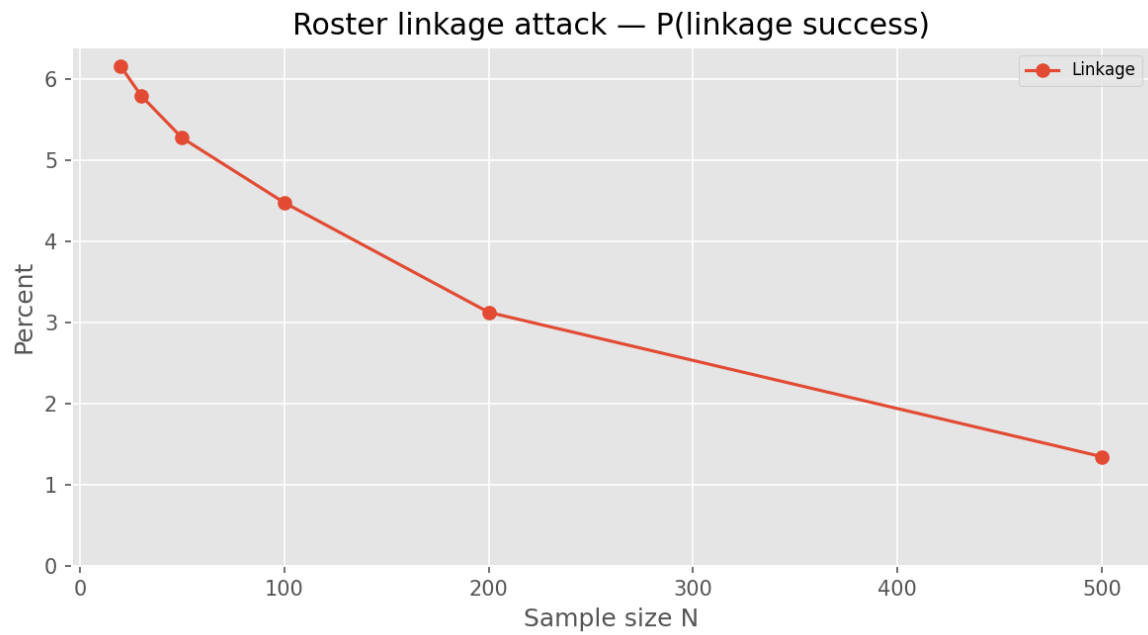


Figure 4: Simplified roster linkage attack: probability of successful match by sample size.

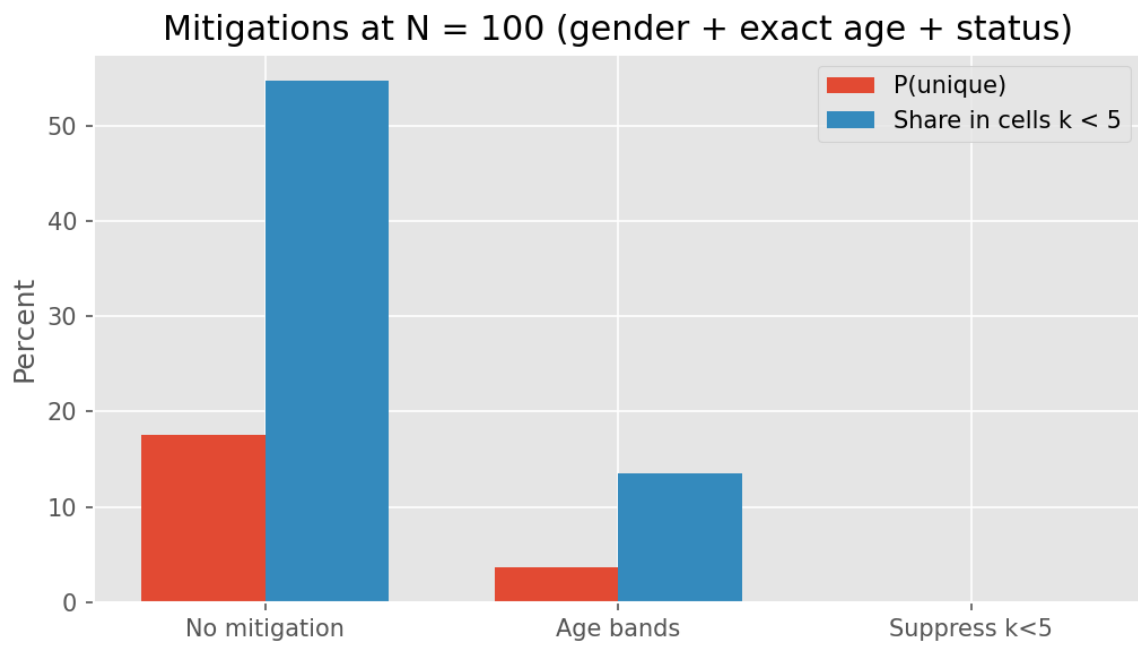


Figure 5: Mitigations at N = 100 for gender + exact age + status.

```
# Optional sanitized microdata for T1
python3 scripts/reid_risk_audit.py \
  --input responses_export.csv \
  --flatten-payload \
  --k-threshold 5 \
  --remediate \
  --output-sanitized responses_sanitized.csv
```

Metric	Meaning
min_k	Smallest group size for a QI combination
groups_below_k	Count of equivalence classes under threshold
pct_unique_rows	Fraction of rows in singleton classes

Remediation patterns implemented in the script:

```
# Coarsen age and dates; strip identifiers; suppress sparse cells
df["age_band"] = pd.cut(df["age"], bins=[18, 25, 35, 45, 55, 65, 200], right=False)
df["created_date"] = pd.to_datetime(df["created_at"], utc=True).dt.floor("D")
drop = [c for c in df.columns if any(h in c.lower() for h in
    ("email", "session_id", "ip", "user_agent", "response_id"))]
df = df.drop(columns=drop, errors="ignore")
```

9 6. Joint responsibility matrix

Risk	IT primary	Research primary
Unauthorized admin/export	RBAC, MFA, logging	Least-privilege accounts
PII in JSON payload	API allowlist (future); WAF	Protocol field hygiene
Temp CSV → R	Secure /tmp, container isolation	SDC before R; no raw <code>saveRDS</code> on sync folders
Demographics + timing mosaic	Consistent proxy IP logging	Coarsen timestamps; limit QIs
Email session bridge	DB permissions on <code>StudyEmailContact</code>	No merge into analysis without IRB
Small cells in public tables	—	Suppress; collapse categories
AI IRB prompts	Vendor contracts, egress	Redact uploads

10 7. Gap summary and recommended actions

Cross-functional (high priority):

1. Payload allowlist at `submit_response` for `email`, `results_email`, `user_agent`, precise `completed_at`, and nonessential external IDs.
2. Wire export sanitization and `get_anonymized_participant_id()` on any CSV leaving Django/Celery.
3. Use `get_client_ip(request)` consistently behind reverse proxies.

4. Coarsen or audit temp CSV before `Rscript` in `run_post_decision_analysis`.

Research (aggregate release path):

1. Enforce `cell n 5` on all external tables and slides.
2. Run `reid_risk_audit.py` before submission when $N < 200$ or 3 QIs are collected.
3. Align goals-refs live survey with age bands and separate email collection.

IT-only:

1. Continue `SECURITY_COMPLIANCE_REMEDIATION_LIST.md` remediations.
2. Encrypt backups containing `Response`; periodic access reviews.
3. Monitor abnormal submission volume per study.

11 8. Conclusion

For studies with 20–500 participants, releasing **aggregates that support inferential claims** is the correct default and aligns with how this lab intends to communicate science externally. IT professionals should treat the application as a **high-trust data processor**: metadata, bridges, and export pipelines are the dominant mosaic vectors. Research teams should treat **every internal CSV as if it could become public**, applying SDC and cell-size discipline before analysis artifacts spread beyond the secure environment. Mermaid diagrams and the audit script in this repository are maintained as **version-controlled source** so IT and research can update controls as protocols evolve.

12 Appendix A — Quasi-identifier inventory

Source	Fields
Response columns	<code>session_id</code> , <code>created_at</code> , <code>ip_address</code> , <code>user_agent</code>
goals-refs payload	<code>demographics.*</code> , <code>prams_participant_id</code> , <code>completed_at</code> , <code>vignette_order_presented</code> , <code>goal_frames</code> , <code>results_email</code>
EXT-AM4 payload	<code>session_id</code> , <code>started_at</code> , <code>completed_at</code> , <code>demographics.age_range</code> , <code>demographics.gender</code> , <code>condition</code>
HR-SJT payload	<code>race_ethnicity</code> , <code>job_level</code> , <code>years_in_hr</code> , <code>education</code> , <code>credentials</code>
Linkage	<code>StudyEmailContact.email + session_id</code> ; <code>StudentDataConsent.email</code>

13 Appendix B — IT operations checklist (one page)

Use this checklist for onboarding, annual review, or pre-production sign-off. Items map to §7 and codebase paths in this repository.

#	Control area	Action	Owner	Done
1	Access	Restrict Django admin to named staff; require MFA for admin and DB access	IT	
2	Access	Verify RBAC: participants cannot self-assign <code>researcher / admin / irb_member</code> (see <code>SECURITY_COMPLIANCE_REMEDIATION_LIST.md</code> H-1)	IT	
3	Access	Confirm IDOR fixes on researcher views (e.g. mark-attendance) return 404, not partial PII	IT	
4	API	Document CSRF/token policy for <code>submit_response</code> and public protocol POST endpoints	IT	
5	API	Confirm only <code>Study.active_approved</code> studies accept submissions	IT / Dev	
6	Network	Standardize client IP: <code>get_client_ip(request)</code> on all response writes behind reverse proxy	Dev	
7	Data store	Encrypt DB backups containing <code>responses</code> , <code>signups</code> , <code>study_email_contacts</code>	IT	
8	Data store	Separate backup/ACL policy for FERPA exports (<code>course_credits_csv</code>) vs research tables	IT	
9	Export pipeline	Review Celery <code>run_post_decision_analysis:</code> temp CSV lifecycle, R host isolation, no shared sync folders	IT	

#	Control area	Action	Owner	Done
10	Export pipeline	Plan: sanitize CSV (strip <code>session_id</code> , IP, raw payload) or run <code>reid_risk_audit.py</code> <code>--remediate</code> before Rscript	Dev / Research	
11	Linkage	Limit DB/user access to <code>StudyEmailContact</code> and <code>StudentDataConsent</code> ; audit cross-table queries	IT	
12	Logging	Centralize <code>AuditLog</code> + app logs; retention policy; alert on bulk CSV export / credit export	IT	
13	Logging	Pass <code>ip_address</code> / <code>user_agent</code> into view-created <code>AuditLog</code> rows (signals lack request context)	Dev	
14	Third party	IRB AI: BAA/DPAs, egress rules, prompt screening per <code>FERPA_AUDIT_REPORT.md</code>	IT / IRB	
15	Monitoring	Alert on abnormal <code>Response</code> create rate per <code>study_id</code> (possible poisoning)	IT	
16	Research attestation	Confirm lab policy: external release = T0 aggregates only; cell $n = 5$	Research PI	
17	Incident	Define procedure if microdata CSV is emailed or uploaded to non-approved storage	IT / IRB	

Sign-off (optional): System owner _____ Date _____ | Security/IT reviewer
 _____ Date _____

14 Appendix C — Document control

Version	Date	Notes
1.0	2026-05-16	Initial report; $N = 20$ –500; aggregate-primary release
1.1	2026-05-16	K-Dense Mermaid integration; Jupyter-free Quarto render
1.2	2026-05-16	Appendix B IT one-page checklist
1.3	2026-05-16	§4.5 simulation narrative, figures, descriptive-table release policy
1.4	2026-05-16	Mermaid % fig-cap fix; §1.2 open science (Castille et al., 2022)
1.5	2026-05-16	HTML theme (<code>mosaic-report.scss</code>); TOC/numbering cleanup

15 References

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